$$
\begin{array}{c|cc}
 & \mathcal{K}^{\#1}_{0^+} & \mathcal{K}^{\#1}_{0^-\,\alpha\beta\chi} \\
\hline
\mathcal{K}^{\#1}_{0^+}{}^{\dagger} & -\frac{3\,Y_1\,k^2}{8} & 0 \\
\mathcal{K}^{\#1}_{0^-}{}^{\dagger\,\alpha\beta\chi} & 0 & 3\,k^2\,\overset{(4)}{\kappa}_{15}
\end{array}
$$

$$
\begin{array}{c|cccc}
 & \mathcal{K}^{\#1}_{1^+\,\alpha\beta} & \mathcal{K}^{\#2}_{1^+\,\alpha\beta} & \mathcal{K}^{\#1}_{1^-\,\alpha} & \mathcal{K}^{\#2}_{1^-\,\alpha} \\
\hline
\mathcal{K}^{\#1}_{1^+}{}^{\dagger\,\alpha\beta} & -\frac{Y_2\,k^2}{8} & \frac{Y_2\,k^2}{4\sqrt{2}} & 0 & 0 \\
\mathcal{K}^{\#2}_{1^+}{}^{\dagger\,\alpha\beta} & \frac{Y_2\,k^2}{4\sqrt{2}} & -\frac{Y_2\,k^2}{4} & 0 & 0 \\
\mathcal{K}^{\#1}_{1^-}{}^{\dagger\,\alpha} & 0 & 0 & -k^2\,\overset{(4)}{\kappa}_2 & -\sqrt{2}\,k^2\,\overset{(4)}{\kappa}_2 \\
\mathcal{K}^{\#2}_{1^-}{}^{\dagger\,\alpha} & 0 & 0 & -\sqrt{2}\,k^2\,\overset{(4)}{\kappa}_2 & -2\,k^2\,\overset{(4)}{\kappa}_2
\end{array}
$$

$$
\begin{array}{c|cc}
 & \mathcal{K}^{\#1}_{2^+\,\alpha\beta} & \mathcal{K}^{\#1}_{2^-\,\alpha\beta\chi} \\
\hline
\mathcal{K}^{\#1}_{2^+}{}^{\dagger\,\alpha\beta} & \frac{9\,Y_3\,k^2}{8} & 0 \\
\mathcal{K}^{\#1}_{2^-}{}^{\dagger\,\alpha\beta\chi} & 0 & 0
\end{array}
$$

$$
\begin{array}{c|cc}
 & \mathcal{J}^{\#1}_{0^+} & \mathcal{J}^{\#1}_{0^-\,\alpha\beta\chi} \\
\hline
\mathcal{J}^{\#1}_{0^+}{}^{\dagger} & \frac{1}{\mathrm{Det}(0^+)} & 0 \\
\mathcal{J}^{\#1}_{0^-}{}^{\dagger\,\alpha\beta\chi} & 0 & \frac{1}{3\,k^2\,\overset{(4)}{\kappa}_{15}}
\end{array}
$$

$$
\begin{array}{c|cccc}
 & \mathcal{J}^{\#1}_{1^+\,\alpha\beta} & \mathcal{J}^{\#2}_{1^+\,\alpha\beta} & \mathcal{J}^{\#1}_{1^-\,\alpha} & \mathcal{J}^{\#2}_{1^-\,\alpha} \\
\hline
\mathcal{J}^{\#1}_{1^+}{}^{\dagger\,\alpha\beta} & \frac{1}{3\,\mathrm{Det}(1^+)} & -\frac{\sqrt{2}}{3\,\mathrm{Det}(1^+)} & 0 & 0 \\
\mathcal{J}^{\#2}_{1^+}{}^{\dagger\,\alpha\beta} & -\frac{\sqrt{2}}{3\,\mathrm{Det}(1^+)} & \frac{2}{3\,\mathrm{Det}(1^+)} & 0 & 0 \\
\mathcal{J}^{\#1}_{1^-}{}^{\dagger\,\alpha} & 0 & 0 & -\frac{1}{9\,k^2\,\overset{(4)}{\kappa}_2} & -\frac{\sqrt{2}}{9\,k^2\,\overset{(4)}{\kappa}_2} \\
\mathcal{J}^{\#2}_{1^-}{}^{\dagger\,\alpha} & 0 & 0 & -\frac{\sqrt{2}}{9\,k^2\,\overset{(4)}{\kappa}_2} & -\frac{2}{9\,k^2\,\overset{(4)}{\kappa}_2}
\end{array}
$$

$$
\begin{array}{c|cc}
 & \mathcal{J}^{\#1}_{2^+\,\alpha\beta} & \mathcal{J}^{\#1}_{2^-\,\alpha\beta\chi} \\
\hline
\mathcal{J}^{\#1}_{2^+}{}^{\dagger\,\alpha\beta} & \frac{1}{\mathrm{Det}(2^+)} & 0 \\
\mathcal{J}^{\#1}_{2^-}{}^{\dagger\,\alpha\beta\chi} & 0 & 0
\end{array}
$$

Abbreviations used in matrices

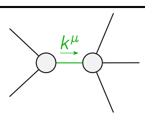
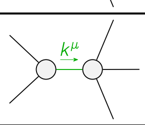
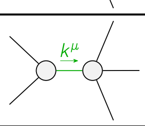
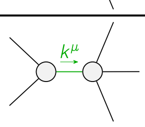
$$Y_1 == 4\,\overset{(4)}{\kappa}_1 - 6\,\overset{(4)}{\kappa}_{15} + \overset{(4)}{\kappa}_2 + 3\,\overset{(4)}{\kappa}_3 \;\&\&\; Y_2 == 2\,\overset{(4)}{\kappa}_{15} + 9\,\overset{(4)}{\kappa}_2 - \overset{(4)}{\kappa}_3 \;\&\&$$
$$Y_3 == 2\,\overset{(4)}{\kappa}_{15} + \overset{(4)}{\kappa}_2 - \overset{(4)}{\kappa}_3 \;\&\&\; \mathrm{Det}(0^+) == -\frac{3}{8}\,k^2\,(4\,\overset{(4)}{\kappa}_1 - 6\,\overset{(4)}{\kappa}_{15} + \overset{(4)}{\kappa}_2 + 3\,\overset{(4)}{\kappa}_3) \;\&\&$$
$$\mathrm{Det}(1^+) == \frac{3}{8}\,k^2\,(-2\,\overset{(4)}{\kappa}_{15} - 9\,\overset{(4)}{\kappa}_2 + \overset{(4)}{\kappa}_3) \;\&\&\; \mathrm{Det}(2^+) == \frac{9}{8}\,k^2\,(2\,\overset{(4)}{\kappa}_{15} + \overset{(4)}{\kappa}_2 - \overset{(4)}{\kappa}_3)$$

Lagrangian

$$
\begin{aligned}
&\overset{(4)}{\kappa}_1\,\partial_\beta\mathcal{K}^{\delta}{}_{\chi\delta}\,\partial^\chi\mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta} + \overset{(4)}{\kappa}_2\,\partial_\chi\mathcal{K}^{\delta}{}_{\beta\delta}\,\partial^\chi\mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta} + \overset{(4)}{\kappa}_3\,\partial_\beta\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\alpha\chi}{}^{\delta} - \\
&3\,\overset{(4)}{\kappa}_{15}\,\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\beta\chi}{}^{\delta} + \frac{9}{2}\,\overset{(4)}{\kappa}_2\,\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\beta\chi}{}^{\delta} - \frac{1}{2}\,\overset{(4)}{\kappa}_3\,\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\beta\chi}{}^{\delta} - \frac{9}{2}\,\overset{(4)}{\kappa}_{15}\,\partial_\beta\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\chi\alpha}{}^{\delta} - \\
&\frac{9}{4}\,\overset{(4)}{\kappa}_2\,\partial_\beta\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\chi\alpha}{}^{\delta} + \frac{5}{4}\,\overset{(4)}{\kappa}_3\,\partial_\beta\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\chi\alpha}{}^{\delta} - 6\,\overset{(4)}{\kappa}_2\,\partial^\chi\mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta}\,\partial_\delta\mathcal{K}_{\chi\beta}{}^{\delta} - \frac{3}{2}\,\overset{(4)}{\kappa}_{15}\,\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}^{\delta}{}_{\beta\chi} - \\
&\frac{9}{4}\,\overset{(4)}{\kappa}_2\,\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}^{\delta}{}_{\beta\chi} + \frac{1}{4}\,\overset{(4)}{\kappa}_3\,\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}^{\delta}{}_{\beta\chi} + \overset{(4)}{\kappa}_{15}\,\partial_\delta\mathcal{K}_{\alpha\beta\chi}\,\partial^\delta\mathcal{K}^{\alpha\beta\chi} - 2\,\overset{(4)}{\kappa}_{15}\,\partial_\delta\mathcal{K}_{\beta\alpha\chi}\,\partial^\delta\mathcal{K}^{\alpha\beta\chi}
\end{aligned}
$$

Added source term(s):	$\mathcal{K}^{\alpha\beta\chi}\;\mathcal{J}_{\alpha\beta\chi}$

Source constraint(s)	# constraint(s)	Covariant form
$\mathcal{J}^{\#1\alpha}_{1^-} == \mathcal{J}^{\#2\alpha}_{1^-}$	3	$\partial_\chi\partial^\alpha\mathcal{J}^{\beta}{}_{\beta}{}^{\chi} + \partial_\chi\partial^\chi\mathcal{J}^{\beta\alpha}{}_{\beta} == 0$
$2\;\mathcal{J}^{\#1\alpha\beta}_{1^+} + \mathcal{J}^{\#2\alpha\beta}_{1^+} == 0$	3	$\partial_\chi\mathcal{J}^{\alpha\beta\chi} + \partial_\chi\mathcal{J}^{\chi\alpha\beta} == \partial_\chi\mathcal{J}^{\beta\alpha\chi}$
$\mathcal{J}^{\#1\alpha\beta\chi}_{2^-} == 0$	5	$3\,\partial_\epsilon\partial_\delta\partial^\chi\partial^\alpha\mathcal{J}^{\delta\beta\epsilon} + 3\,\partial_\epsilon\partial^\epsilon\partial^\chi\partial^\alpha\mathcal{J}^{\delta\beta}{}_{\delta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\beta\mathcal{J}^{\alpha\chi\delta} + 4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\beta\mathcal{J}^{\chi\alpha\delta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\beta\mathcal{J}^{\delta\alpha\chi} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\chi\mathcal{J}^{\beta\alpha\delta} + 4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\chi\mathcal{J}^{\delta\alpha\beta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\delta\mathcal{J}^{\alpha\beta\chi} + 3\,\eta^{\beta\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial^\alpha\mathcal{J}^{\delta}{}_{\delta}{}^{\epsilon} + 3\,\eta^{\alpha\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial_\delta\mathcal{J}^{\delta\beta\epsilon} + 3\,\eta^{\beta\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial^\epsilon\mathcal{J}^{\delta\alpha}{}_{\delta} == 3\,\partial_\epsilon\partial_\delta\partial^\chi\partial^\beta\mathcal{J}^{\delta\alpha\epsilon} + 3\,\partial_\epsilon\partial^\epsilon\partial^\chi\partial^\beta\mathcal{J}^{\delta\alpha}{}_{\delta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\alpha\mathcal{J}^{\beta\chi\delta} + 4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\alpha\mathcal{J}^{\chi\beta\delta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\alpha\mathcal{J}^{\delta\beta\chi} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\chi\mathcal{J}^{\alpha\beta\delta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\delta\mathcal{J}^{\beta\alpha\chi} + 4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\delta\mathcal{J}^{\chi\alpha\beta} + 3\,\eta^{\alpha\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial^\beta\mathcal{J}^{\delta}{}_{\delta}{}^{\epsilon} + 3\,\eta^{\beta\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial_\delta\mathcal{J}^{\delta\alpha\epsilon} + 3\,\eta^{\alpha\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial^\epsilon\mathcal{J}^{\delta\beta}{}_{\delta}$
Total # constraint(s):	11	

Resolved pole(s)	# polarization(s)	Square mass	Residue
	1	0	$\frac{4\,\overset{(4)}{\kappa}_{15}{}^2+20\,\overset{(4)}{\kappa}_{15}\,\overset{(4)}{\kappa}_2+57\,\overset{(4)}{\kappa}_2{}^2-4\,\overset{(4)}{\kappa}_{15}\,\overset{(4)}{\kappa}_3-10\,\overset{(4)}{\kappa}_2\,\overset{(4)}{\kappa}_3+\overset{(4)}{\kappa}_3{}^2}{\overset{(4)}{\kappa}_2\,(2\,\overset{(4)}{\kappa}_{15}+\overset{(4)}{\kappa}_2-\overset{(4)}{\kappa}_3)\,(2\,\overset{(4)}{\kappa}_{15}+9\,\overset{(4)}{\kappa}_2-\overset{(4)}{\kappa}_3)}$
	1	0	$\frac{36\,\overset{(4)}{\kappa}_{15}{}^2+97\,\overset{(4)}{\kappa}_2{}^2+4\,\overset{(4)}{\kappa}_{15}\,(29\,\overset{(4)}{\kappa}_2-9\,\overset{(4)}{\kappa}_3)-58\,\overset{(4)}{\kappa}_2\,\overset{(4)}{\kappa}_3+9\,\overset{(4)}{\kappa}_3{}^2}{\overset{(4)}{\kappa}_2\,(2\,\overset{(4)}{\kappa}_{15}+\overset{(4)}{\kappa}_2-\overset{(4)}{\kappa}_3)\,(2\,\overset{(4)}{\kappa}_{15}+9\,\overset{(4)}{\kappa}_2-\overset{(4)}{\kappa}_3)}$
	1	0	$-\frac{48\,\overset{(4)}{\kappa}_{15}\,\overset{(4)}{\kappa}_2+216\,\overset{(4)}{\kappa}_2{}^2-24\,\overset{(4)}{\kappa}_2\,\overset{(4)}{\kappa}_3+\sqrt{(-2\,\overset{(4)}{\kappa}_{15}-9\,\overset{(4)}{\kappa}_2+\overset{(4)}{\kappa}_3)^2\,(324\,\overset{(4)}{\kappa}_{15}{}^2+900\,\overset{(4)}{\kappa}_{15}\,\overset{(4)}{\kappa}_2+1841\,\overset{(4)}{\kappa}_2{}^2-324\,\overset{(4)}{\kappa}_{15}\,\overset{(4)}{\kappa}_3-450\,\overset{(4)}{\kappa}_2\,\overset{(4)}{\kappa}_3+81\,\overset{(4)}{\kappa}_3{}^2)}}{\overset{(4)}{\kappa}_2\,(2\,\overset{(4)}{\kappa}_{15}+\overset{(4)}{\kappa}_2-\overset{(4)}{\kappa}_3)\,(2\,\overset{(4)}{\kappa}_{15}+9\,\overset{(4)}{\kappa}_2-\overset{(4)}{\kappa}_3)}$
	1	0	$\frac{-48\,\overset{(4)}{\kappa}_{15}\,\overset{(4)}{\kappa}_2-216\,\overset{(4)}{\kappa}_2{}^2+24\,\overset{(4)}{\kappa}_2\,\overset{(4)}{\kappa}_3+\sqrt{(-2\,\overset{(4)}{\kappa}_{15}-9\,\overset{(4)}{\kappa}_2+\overset{(4)}{\kappa}_3)^2\,(324\,\overset{(4)}{\kappa}_{15}{}^2+900\,\overset{(4)}{\kappa}_{15}\,\overset{(4)}{\kappa}_2+1841\,\overset{(4)}{\kappa}_2{}^2-324\,\overset{(4)}{\kappa}_{15}\,\overset{(4)}{\kappa}_3-450\,\overset{(4)}{\kappa}_2\,\overset{(4)}{\kappa}_3+81\,\overset{(4)}{\kappa}_3{}^2)}}{\overset{(4)}{\kappa}_2\,(2\,\overset{(4)}{\kappa}_{15}+\overset{(4)}{\kappa}_2-\overset{(4)}{\kappa}_3)\,(2\,\overset{(4)}{\kappa}_{15}+9\,\overset{(4)}{\kappa}_2-\overset{(4)}{\kappa}_3)}$
Resolved unitarity condition(s):		(Demonstrably impossible)	